

K -theoretic Catalan functions

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Overview

- ① Schubert calculus
- ② Catalan functions
- ③ K-theoretic Catalan functions

Overview of Schubert Calculus Combinatorics

Geometric problem

Find $c_{\lambda\mu}^\nu = \#$ of points in intersection of subvarieties in a variety X .

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Special basis of polynomials $\{f_\lambda\}$ such that $f_\lambda \cdot f_\mu = \sum_\nu c_{\lambda\mu}^\nu f_\nu$

Classical Schubert Calculus Example

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Find $c_{\lambda\mu}^\nu = \#$ of points in intersection of Schubert varieties $\{X_\lambda\}_{\lambda \subseteq (n^m)}$ in variety $X = \text{Gr}(m, n)$.

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Representatives

Special basis of Schur polynomials $\{s_\lambda\}$ such that $s_\lambda \cdot s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$ for combinatorially understood Littlewood-Richardson coefficients $c_{\lambda\mu}^\nu$.

Schur polynomials and raising operators

- Complete homogeneous symmetric function: for $r \in \mathbb{Z}$,
$$h_r = \sum_{i_1 \leq \dots \leq i_r} x_{i_1} \cdots x_{i_r}.$$

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- Raising operators $R_{i,j}(h_\lambda) = h_{\lambda + \epsilon_i - \epsilon_j}$

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- Schur function $s_\lambda = \prod_{i < j} (1 - R_{ij}) h_\lambda$ (Jacobi-Trudi)

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Quantum cohomology of flag variety	Quantum Schuberts
(Co)homology of Types BCD Grassmannian	Schur- P and Q functions
(Co)homology of affine Grassmannian	(dual) k -Schur functions
K -theory of Grassmannian	Grothendieck polynomials
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K -theory and K -homology of the affine Grassmannian

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Focus

K -theory and K -homology of the affine Grassmannian

Simultaneously generalizes K -theory of Grassmannian and (co)homology of affine Grassmannian.

What is known?

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- ① K -theory classes of Grassmannian (not affine!) represented by “Grothendieck polynomials.” We are interested in their dual:

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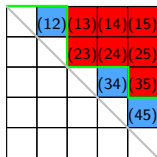
- ② Homology classes of affine Grassmannian represented by k -Schur functions ($t = 1$).
- ③ (Lam et al., 2010) leave open the question: what is a direct formulation of the K -homology representatives of the affine Grassmannian (K - k -Schur functions)?

Goal

Identify K - k -Schur functions in explicit (simple) terms amenable to calculation and proofs.

Root Ideals

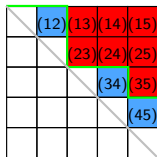
A root ideal Ψ of type $A_{\ell-1}$ positive roots: given by Dyck path (lattice path above diagonal).



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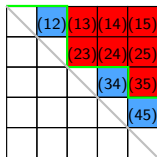
Catalan Function (Chen, 2010; Panyushev, 2010; Blasiak et al., 2019)

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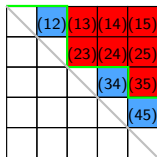
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- $\Psi = \text{all roots} \implies H(\Psi; \gamma) = h_{\gamma}$

k -Schur root ideal for λ

For $k \in \mathbb{Z}_{\geq 0}$ and $\lambda = (\lambda_1 \geq \dots \geq \lambda_\ell) \in \mathbb{Z}^\ell$,

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$$\Delta^4(3, 3, 2, 2, 1, 1) =$$

3					
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		2			
			2		
				1	
					1

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k -Schur is a Catalan function (Blasiak et al., 2019).

For partition λ with $\lambda_1 \leq k$,

$$s_\lambda^{(k)} = H(\Delta^k(\lambda); \lambda).$$

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Remark

(Blasiak et al., 2019) show results for k -Schur functions with parameter t , but $t = 1$ specialization is necessary for Schubert calculus.

Lowering Operators

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- Extra ingredient: lowering operators $L_j(h_\lambda) = h_{\lambda - \epsilon_j}$

$$L_3 \left(\begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & & & \\ \hline \square & & & \\ \hline \end{array} \right) = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \square & & \\ \hline \end{array} \quad L_1 \left(\begin{array}{|c|c|c|c|} \hline \square & \square & \square & \color{red}\square \\ \hline \square & & & \\ \hline \square & & & \\ \hline \end{array} \right) = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \square & \\ \hline \end{array}$$

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$$K(\Psi; \mathcal{L}; \gamma) := \prod_{(i,j) \in \mathcal{L}} (1 - L_j) \prod_{(i,j) \in \Delta_\ell^+ \setminus \Psi} (1 - R_{ij}) k_\gamma$$

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Example

non-roots of Ψ in blue, roots of $\mathcal{L}$ marked with •

	(12)		•	•
			•	•
		(34)		
			(45)	

$$\begin{aligned} K(\Psi; \mathcal{L}; 54332) \\ &= (1 - L_4)^2 (1 - L_5)^2 \\ &\cdot (1 - R_{12})(1 - R_{34})(1 - R_{45}) k_{54332} \end{aligned}$$

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Example

$$g_{332111}^{(4)} = \begin{array}{|c|c|c|c|c|c|c|} \hline 3 & \text{blue} & & \bullet & \bullet & \bullet & \\ \hline & 3 & \text{blue} & & \bullet & \bullet & \\ \hline & & & 2 & \text{blue} & \text{blue} & \\ \hline & & & & 1 & \text{blue} & \text{blue} \\ \hline & & & & & 1 & \text{blue} \\ \hline & & & & & & 1 \\ \hline \end{array} \quad \Delta^+ \setminus \Psi = \Delta_6^+ \setminus \Delta^{(4)}(332111), \mathcal{L} = \Delta^{(5)}(332111)$$

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Property and Further Work

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Theorem (Blasiak-Morse-S., 2022)

The $g_{\lambda}^{(k)}$ “branching coefficients” are alternating by degree, i.e. the $b_{\lambda\mu}^{(k)}$ in

$$g_{\lambda}^{(k)} = \sum_{\mu} b_{\lambda\mu}^{(k)} g_{\mu}^{(k+1)}$$

satisfy $(-1)^{|\lambda|-|\mu|} b_{\lambda\mu}^{(k)} \in \mathbb{Z}_{\geq 0}$.

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Under the Peterson Isomorphism, the “quantum Grothendieck polynomials” $\mathfrak{G}_w(z; Q)$ get sent to “closed K - k -Schur functions”, $\mathfrak{g}_\lambda^{(k)} = K(\Delta^{(k)}; \Delta^{(k)}; \lambda)$ with suitable localization.

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Proved using “Katalan function description.”

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$$G_{1^r}^{\perp} g_{\lambda}^{(k)} = \sum_{\mu} ?? g_{\mu}^{(k)} \iff G_{1^r} G_{\mu}^{(k)} = \sum_{\lambda} ?? G_{\lambda}^{(k)}, \quad 1 \leq r \leq k.$$

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- 3 Combinatorially describe $g_\lambda^{(k)} = \sum_\mu ?? s_\mu^{(k)}$.
- 4 Answer same questions for “closed K - k -Schur’s.”

Other results using Catalan function methods

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- ② Also provides methods to prove “Schur positivity” of families of symmetric functions. (Blasiak-Morse-Pun 2020, Blasiak-Haiman-Morse-Pun-S. 2021+)
- ③ New formulas for Macdonald polynomials using raising operators (Blasiak-Haiman-Morse-Pun-S.)

Thank you!

Blasiak, Jonah, Mark Haiman, Jennifer Morse, Anna Pun, and George H. Seelinger. 2023. *A Shuffle Theorem for Paths Under Any Line*, Forum of Mathematics, Pi **11**.

———. 2021. *Dens, Nests and the Loehr-Warrington Conjecture*, arXiv.

Blasiak, Jonah, Jennifer Morse, and Anna Pun. 2020. *Demazure Crystals and the Schur Positivity of Catalan Functions*, arXiv.

Blasiak, Jonah, Jennifer Morse, Anna Pun, and Daniel Summers. 2019. *Catalan Functions and k -Schur Positivity*, Journal of the AMS.

Blasiak, Jonah, Jennifer Morse, and George H. Seelinger. 2022. *K -Theoretic Catalan Functions*, Advances in Mathematics **404**.

Chen, Li-Chung. 2010. *Skew-linked partitions and a representation theoretic model for k -Schur functions*, Ph.D. thesis.

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Ikeda, Takeshi, Shinsuke Iwao, and Toshiaki Maeno. 2020. *Peterson Isomorphism in K -theory and Relativistic Toda Lattice*, International Mathematics Research Notices **19**.

Ikeda, Takeshi, Shinsuke Iwao, and Satoshi Naito. 2022. *Closed KK -Schur Katalan Functions as KK -Homology Schubert Representatives of the Affine Grassmannian*, arXiv.

Lam, Thomas, Anne Schilling, and Mark Shimozono. 2010. *K -theory Schubert calculus of the affine Grassmannian*, Compositio Math. **146**, 811–852.

Morse, Jennifer. 2011. *Combinatorics of the K -theory of affine Grassmannians*, Advances in Mathematics.

Panyushev, Dmitri I. 2010. *Generalised Kostka-Foulkes polynomials and cohomology of line bundles on homogeneous vector bundles*, Selecta Math. (N.S.) **16**, no. 2, 315–342.